

UAV MOTION PLANNING HANDOUT

When “Smooth and Fast” Is a Theorem

Minimum-snap trajectories in a fixed convex corridor

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Claim supported by slide 13. Once the flight time, polynomial degree, segment allocation, and a *fixed convex safe corridor* are chosen, minimum-snap trajectory generation can be written as a convex quadratic program (QP). Its objective is convex, its standard constraints are affine, and therefore every local minimizer is global. This theorem does not cover arbitrary collision constraints or free-time optimization.

1 Piecewise-polynomial trajectory

Let the trajectory have M segments. Segment i occupies $t \in [0, T_i]$ and, in one spatial coordinate, is

$$p_i(t) = \sum_{k=0}^K a_{ik} t^k = \phi(t)^T \mathbf{a}_i, \quad \phi(t) = [1, t, \dots, t^K]^T.$$

The r th derivative is linear in the coefficients:

$$p_i^{(r)}(t) = \phi^{(r)}(t)^T \mathbf{a}_i, \quad \frac{d^r}{dt^r} t^k = \begin{cases} \frac{k!}{(k-r)!} t^{k-r}, & k \geq r, \\ 0, & k < r. \end{cases}$$

Stack every segment and coordinate into one vector $\mathbf{x}$. The map from $\mathbf{x}$ to position, velocity, acceleration, jerk, or snap at any fixed time is affine (linear when no offset is present).

2 Integrated squared snap is a PSD quadratic

For one segment the snap cost is

$$J_i(\mathbf{a}_i) = \int_0^{T_i} (p_i^{(4)}(t))^2 dt = \int_0^{T_i} \mathbf{a}_i^T \phi^{(4)}(t) \phi^{(4)}(t)^T \mathbf{a}_i dt = \mathbf{a}_i^T Q_i \mathbf{a}_i,$$

where

$$(Q_i)_{k\ell} = \begin{cases} \frac{k!}{(k-4)!} \frac{\ell!}{(\ell-4)!} \frac{T_i^{k+\ell-7}}{k+\ell-7}, & k, \ell \geq 4, \\ 0, & \text{otherwise.} \end{cases}$$

For every vector z ,

$$z^T Q_i z = \int_0^{T_i} (\phi^{(4)}(t)^T z)^2 dt \geq 0,$$

so $Q_i \succeq 0$. Summing segments and the three Cartesian coordinates gives

$$J(\mathbf{x}) = \mathbf{x}^T Q \mathbf{x}, \quad Q = \text{blkdiag}(Q_1, \dots, Q_M) \succeq 0.$$

Adding positive weights on other squared derivatives preserves positive semidefiniteness. A positive-semidefinite Hessian $2Q$ proves convexity; it need not be strictly positive definite because low-order polynomials have zero snap.

3 Boundary and continuity constraints are affine

Prescribed derivatives at the start or goal have the form

$$p_i^{(r)}(\tau) = d \iff \phi^{(r)}(\tau)^T \mathbf{a}_i = d.$$

Continuity through derivative order s at a junction is

$$p_i^{(r)}(T_i) = p_{i+1}^{(r)}(0), \quad r = 0, \dots, s,$$

or

$$\phi^{(r)}(T_i)^T \mathbf{a}_i - \phi^{(r)}(0)^T \mathbf{a}_{i+1} = 0.$$

All endpoint and junction conditions therefore stack into

$$A\mathbf{x} = b.$$

Enforcing $r = 0, 1, 2, 3$ makes position, velocity, acceleration, and jerk continuous. The trajectory is then C^3 across segment boundaries. Penalizing snap controls roughness inside segments, while continuity is supplied explicitly by the equalities.

4 A fixed convex corridor gives linear inequalities

Suppose segment i must remain inside a convex polytope

$$\mathcal{P}_i = \{p \in \mathbb{R}^3 : H_i p \leq h_i\}.$$

Pointwise constraints at fixed sample times are affine in $\mathbf{x}$. A stronger whole-segment certificate is available in a Bernstein basis. Write

$$p_i(s) = \sum_{j=0}^K c_{ij} B_{j,K}(s), \quad s \in [0, 1],$$

where $B_{j,K}(s) \geq 0$ and $\sum_j B_{j,K}(s) = 1$. Thus $p_i(s)$ lies in the convex hull of its control points. If

$$H_i c_{ij} \leq h_i \quad \text{for every } j,$$

then convexity implies $H_i p_i(s) \leq h_i$ for every $s \in [0, 1]$. Since the control points are a linear transformation of polynomial coefficients, these are linear inequalities

$$G\mathbf{x} \leq g.$$

Derivative control points similarly impose conservative componentwise velocity, acceleration, jerk, or thrust-related bounds. Exact Euclidean-norm bounds produce second-order cone constraints, which remain convex.

5 The resulting convex program

With fixed times and corridors, the standard problem is

$$\min_{\mathbf{x}} \frac{1}{2} \mathbf{x}^T H \mathbf{x} + f^T \mathbf{x} \quad \text{s.t.} \quad A\mathbf{x} = b, \quad G\mathbf{x} \leq g, \quad H \succeq 0.$$

Its feasible set is convex: affine equalities define an affine set and affine inequalities define half-spaces. The objective is convex because $H \succeq 0$.

Theorem 1 (Every local minimum is global). *For a convex function F on a convex feasible set $\mathcal{X}$, every locally minimizing feasible point x^* is globally minimizing.*

Proof. If a feasible y had $F(y) < F(x^*)$, then $x_\theta = (1 - \theta)x^* + \theta y$ is feasible. Convexity gives

$$F(x_\theta) \leq (1 - \theta)F(x^*) + \theta F(y) < F(x^*)$$

for every $\theta \in (0, 1]$. Taking θ arbitrarily small contradicts local optimality. $\square$

6 KKT conditions are sufficient

The Lagrangian is

$$\mathcal{L}(\mathbf{x}, \lambda, \mu) = \frac{1}{2}\mathbf{x}^T H \mathbf{x} + \mathbf{f}^T \mathbf{x} + \lambda^T (A\mathbf{x} - \mathbf{b}) + \mu^T (G\mathbf{x} - \mathbf{g}).$$

The KKT conditions are

$$\begin{aligned} H\mathbf{x}^* + \mathbf{f} + A^T \lambda^* + G^T \mu^* &= 0 && \text{stationarity,} \\ A\mathbf{x}^* = \mathbf{b}, \quad G\mathbf{x}^* &\leq \mathbf{g} && \text{primal feasibility,} \\ \mu^* &\geq 0 && \text{dual feasibility,} \\ \mu_j^* (G_j \mathbf{x}^* - g_j) &= 0 && \text{complementarity.} \end{aligned}$$

For this convex QP, any feasible primal-dual point satisfying KKT is a global optimum. Under a constraint qualification such as Slater's condition, KKT is also necessary. If H is positive definite on the feasible tangent space, the primal optimum is unique.

7 Why it can be computationally fast

Each cost block touches one segment; a continuity row touches only two neighboring segments. Hence H , A , and usually G are sparse and banded. Equality-only problems solve the sparse linear KKT system

$$\begin{bmatrix} H & A^T \\ A & 0 \end{bmatrix} \begin{bmatrix} \mathbf{x} \\ \lambda \end{bmatrix} = \begin{bmatrix} -\mathbf{f} \\ \mathbf{b} \end{bmatrix}.$$

Interior-point and operator-splitting QP methods exploit the same sparsity. “Fast” therefore means that a structured convex program can be solved reliably and efficiently; it is not a universal runtime guarantee independent of problem size, conditioning, or requested accuracy.

8 Exactly where the theorem stops

Change	Consequence
Fixed convex corridor	Convex linear or conic containment constraints.
Raw obstacle avoidance $p(t) \notin \mathcal{O}$	Complement of an obstacle is generally nonconvex.
Fixed segment times T_i	Coefficients enter a convex quadratic objective.
Optimize times jointly	Powers of T_i couple nonlinearly to coefficients; generally nonconvex.
Linearized dynamics	May remain convex but is an approximation.
Full nonlinear UAV dynamics	Generally a nonlinear, nonconvex optimal-control problem.

9 Takeaway

The slide's positive claim rests on a precise chain:

$\begin{aligned}\text{polynomial coefficients} &\Rightarrow J = \mathbf{x}^T Q \mathbf{x}, \quad Q \succeq 0, \\ \text{fixed convex corridor} &\Rightarrow A \mathbf{x} = b, \quad G \mathbf{x} \leq g, \\ \text{convex QP} &\Rightarrow \text{every local minimum is global.}\end{aligned}$
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Smoothness follows from the chosen polynomial class and explicit derivative-continuity constraints. Dynamics awareness follows only for the bounds actually imposed. Computational speed comes from convexity and sparse local coupling—not from optimization in general.